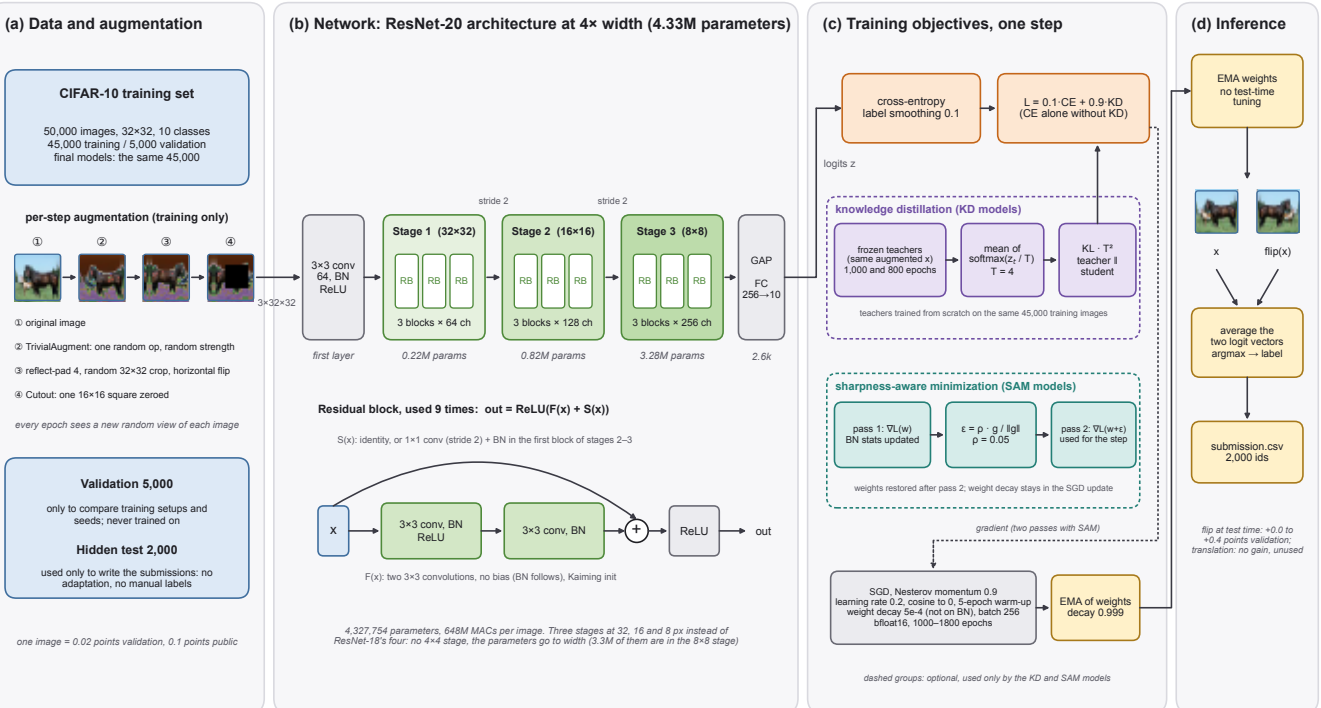

(a) Data and augmentation
CIFAR-10 training set
50,000 images, 32×32, 10 classes
45,000 training / 5,000 validation
final models: the same 45,000
per-step augmentation (training only)
① original image
② TrivialAugment: one random op, random strength
③ reflect-pad 4, random 32×32 crop, horizontal flip
④ Cutout: one 16×16 square zeroed
every epoch sees a new random view of each image
Validation 5,000
only to compare training setups and seeds; never trained on
Hidden test 2,000
used only to write the submissions: no adaptation, no manual labels
one image = 0.02 points validation, 0.1 points public
3×32×32
(b) Network: ResNet-20 architecture at 4× width (4.33M parameters)
3×3 conv 64, BN ReLU
first layer
stride 2
Stage 1 (32×32)
RB RB RB
3 blocks × 64 ch
0.22M params
stride 2
Stage 2 (16×16)
RB RB RB
3 blocks × 128 ch
0.82M params
Stage 3 (8×8)
RB RB RB
3 blocks × 256 ch
3.28M params
GAP
FC 256→10
2.6k
Residual block, used 9 times: out = ReLU(F(x) + S(x))
S(x): identity, or 1×1 conv (stride 2) + BN in the first block of stages 2–3
x
3×3 conv, BN ReLU
3×3 conv, BN
+
ReLU
out
F(x): two 3×3 convolutions, no bias (BN follows), Kaiming init
4,327,754 parameters, 648M MACs per image. Three stages at 32, 16 and 8 px instead of ResNet-18's four: no 4×4 stage, the parameters go to width (3.3M of them are in the 8×8 stage)
(c) Training objectives, one step
cross-entropy label smoothing 0.1
L = 0.1·CE + 0.9·KD (CE alone without KD)
logits z
knowledge distillation (KD models)
frozen teachers (same augmented x) 1,000 and 800 epochs
mean of softmax(z_t / T) T = 4
KL · T² teacher ‖ student
teachers trained from scratch on the same 45,000 training images
sharpness-aware minimization (SAM models)
pass 1: ∇L(w) BN stats updated
ε = ρ · g / ‖g‖ ρ = 0.05
pass 2: ∇L(w+ε) used for the step
weights restored after pass 2; weight decay stays in the SGD update
gradient (two passes with SAM)
SGD, Nesterov momentum 0.9 learning rate 0.2, cosine to 0, 5-epoch warm-up weight decay 5e-4 (not on BN), batch 256 bfloat16, 1000–1800 epochs
EMA of weights decay 0.999
dashed groups: optional, used only by the KD and SAM models
(d) Inference
EMA weights no test-time tuning
x
flip(x)
average the two logit vectors argmax → label
submission.csv 2,000 ids
flip at test time: +0.0 to +0.4 points validation; translation: no gain, unused